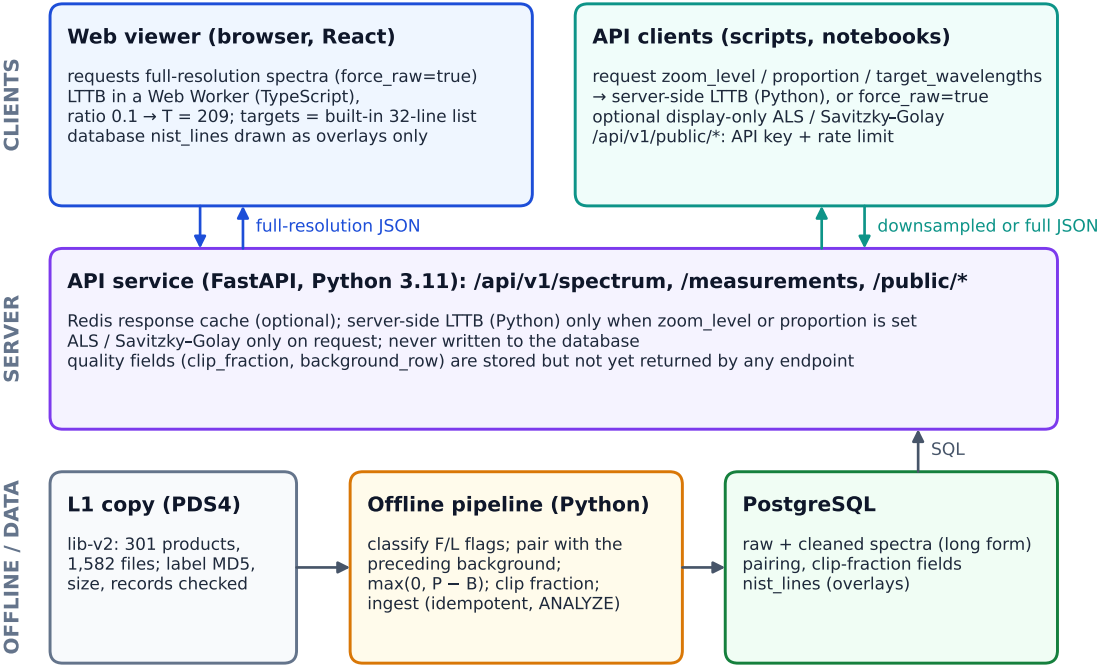
CLIENTS
Web viewer (browser, React)
requests full-resolution spectra (force_raw=true)
LTTB in a Web Worker (TypeScript),
ratio 0.1 → T = 209; targets = built-in 32-line list
database nist_lines drawn as overlays only
API clients (scripts, notebooks)
request zoom_level / proportion / target_wavelengths
→ server-side LTTB (Python), or force_raw=true
optional display-only ALS / Savitzky–Golay
/api/v1/public/*: API key + rate limit
full-resolution JSON
downsampled or full JSON
SERVER
API service (FastAPI, Python 3.11): /api/v1/spectrum, /measurements, /public/*
Redis response cache (optional); server-side LTTB (Python) only when zoom_level or proportion is set
ALS / Savitzky–Golay only on request; never written to the database
quality fields (clip_fraction, background_row) are stored but not yet returned by any endpoint
SQL
OFFLINE / DATA
L1 copy (PDS4)
lib-v2: 301 products,
1,582 files; label MD5,
size, records checked
Offline pipeline (Python)
classify F/L flags; pair with the
preceding background;
max(0, P − B); clip fraction;
ingest (idempotent, ANALYZE)
PostgreSQL
raw + cleaned spectra (long form)
pairing, clip-fraction fields
nist_lines (overlays)